Microsoft AI Engineer Program via Simplilearn

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Sales Forecasting

AIML Capstone — Capstone 3 Writeup

Problem

Fresh Analytics wants a read on what demand across their restaurants' menu items is going to look like, so they can plan production, staffing, and sales ahead of time instead of reacting after the fact. The deliverable is a forecast of daily revenue for the year ahead.

Data

Three files: `resturants.csv` (6 restaurants), `items.csv` (100 items with calories and cost, each tied to a store), `sales.csv` (109,600 rows, a complete 100-item by 1,096-day grid running 2019-01-01 through 2021-12-31, `item_count = 0` on days nothing sold).

What I did

Preliminary check. No nulls, no duplicate rows, nothing negative in any of the three files. `sales.price` matches `items.cost` on every single row, so prices are fixed per item rather than moving day to day. 78.6% of the sales rows are zero-count, which just reflects that most items don't sell every single day on a complete grid like this — not a data quality issue. Merged the three tables on `item_id` and `store_id` and added a `revenue` column (`price × item_count`).

EDA. - Daily revenue across all restaurants averages about $5,843/day (std $2,208), with a 30-day rolling average making the underlying trend visible under the daily noise. - Clear seasonality by quarter. Q2 and Q3 run ahead of Q1 and Q4 (roughly $643K and $634K per quarter vs. $423K and

$435K, averaged over the three years). - Bob's Diner accounts for almost all of the revenue: $6.34M of the roughly $6.42M total across three years. The other five restaurants combined don't come close. It's a volume story — about 627 units/day at Bob's Diner vs. 1-2 units/day everywhere else. - The top-selling item company-wide is the Strawberry Smoothy at 236,337 units, all sold at Bob's Diner, followed by the Frozen Milky Smoothy at 103,263 units. Also pulled the top seller for each individual restaurant. - At the company level, the highest-volume store and the highest-revenue store are the same one (Bob's Diner). The pipeline checks them separately since that overlap isn't guaranteed in general; a store selling a lot of cheap items can still lose to a store selling less at a higher price. - Found the most expensive item at each restaurant along with its calorie count, which is useful for menu planning.

Forecasting. Engineered date-based features: `year`, `month`, `quarter`, `day_of_month`, `day_of_week`, `day_of_year`, `is_weekend`. Target is total daily revenue across all restaurants rather than per-item counts, since the business question here is company-level planning, and per-item forecasting on a panel this sparse would mostly add noise. Held out the last 6 months (2021-07-01 through 2021-12-31, 184 days) as test, trained on the preceding 912 days. Compared three models:

Model	Test RMSE
Linear Regression	$2,234.38
Random Forest	$847.99
XGBoost	**$797.35**

XGBoost won by a clear margin. Makes sense — the relationship between calendar features and revenue isn't linear (the weekend effect isn't the same size in every month, for example), and tree models handle that kind of interaction better than a straight-line fit. Refit XGBoost on the full three years and used it to forecast the next 365 days.

Results

Metric	Value
Rows analyzed (merged panel)	109,600
Data quality issues found	None (0 nulls, 0 duplicates, 0 negative values)
Revenue leader	Bob's Diner ($6.34M of ~$6.42M total, 3-year)
Top-selling item	Strawberry Smoothy (236,337 units)

Metric	Value
Best forecasting model	XGBoost (Test RMSE $797.35 vs. Linear Regression $2,234.38)
Forecast total, next 365 days	$2,396,376
Actual total, most recent year (2021)	$2,398,295

Notes

The forecast for the next year ($2,396,376) comes out within 0.1% of what 2021 actually did ($2,398,295), which is a decent sanity check given the revenue series doesn't show much growth or decline year over year, mostly just the same seasonal pattern repeating. XGBoost's error ($797 RMSE) works out to about 13.6% of average daily revenue, which is reasonable for a model that only has calendar features to work with and nothing on promotions, weather, or local events. This covers the assignment: merge and inspect the data, run EDA from a few different angles (time, restaurant, item, price/calorie), then compare three forecasting models and use the winner for the actual forecast.